Theory of Automata
CS 3005
Lecture 24

Mahzaib Younas

Lecturer, Department of Computer Science

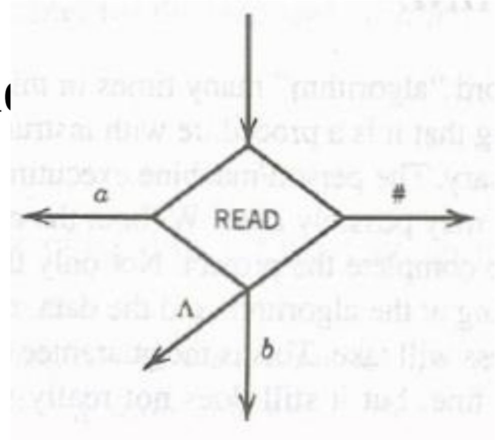
FAST NUCES CFD

POST Machine

- **A Post machine** denoted by **PM** is a collection of five things
  1. The alphabet Σ plus the special symbol #. We generally use Σ = { a, b}.
  2. **A** linear storage location (a place where a string of symbols is kept called the **STORE or QUEUE,** which initially contains the input string.
     1. This location can be read by which we mean the leftmost character can be removed for inspection.
     2. The STORE can also be added to, which means a new character can be concatenated onto the right of whatever is there already.
     3. We allow for the possibility that characters not in Σ can be used in the **STORE,** characters from an alphabet Γ called the store alphabet.

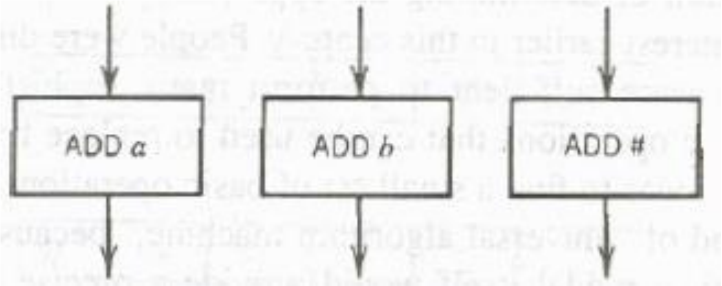
POST Machine Contd.

3. READ states, for exampl



PMs **are** deterministic, so no two edges from the **READ** have the same

4. ADD states:

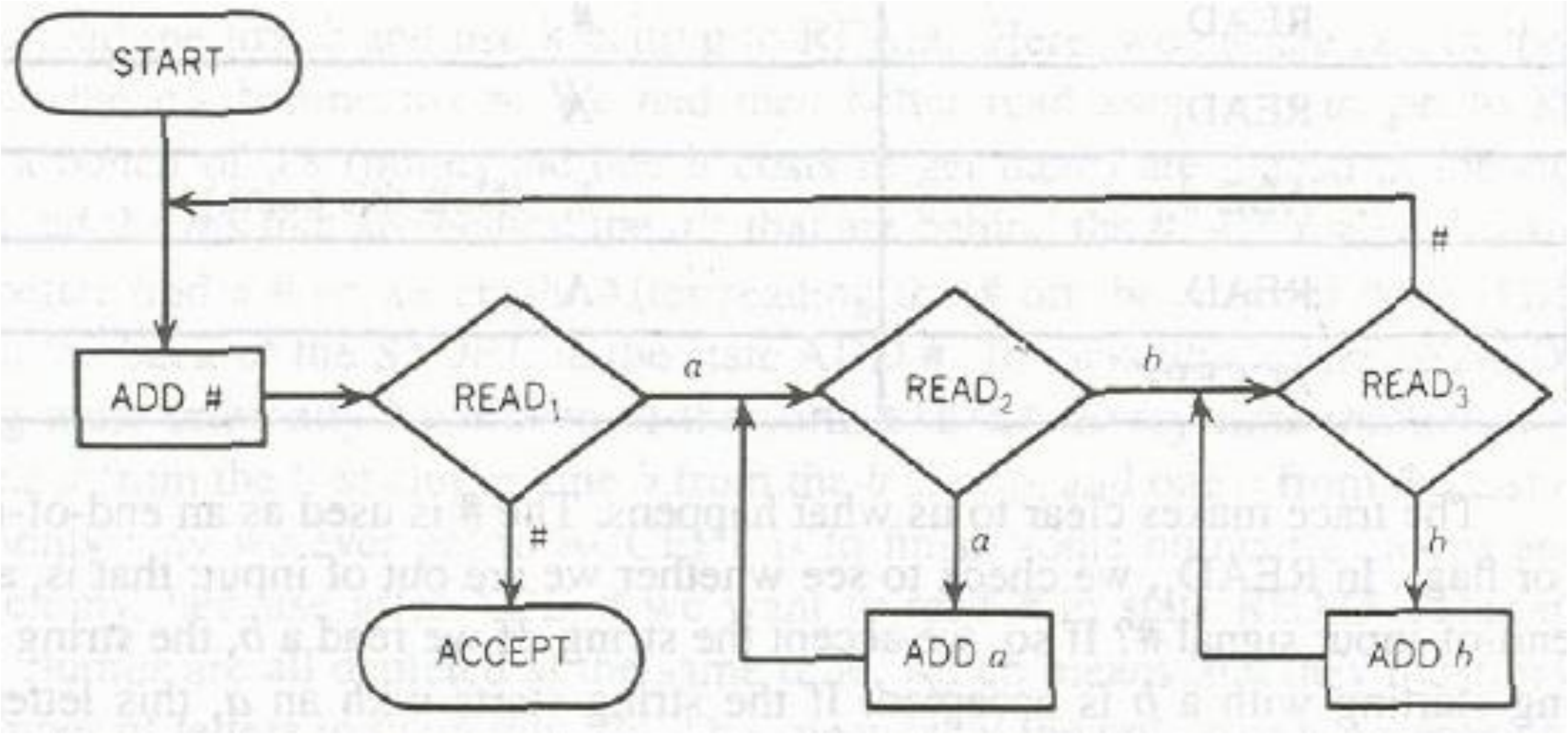


POST Machine Contd.

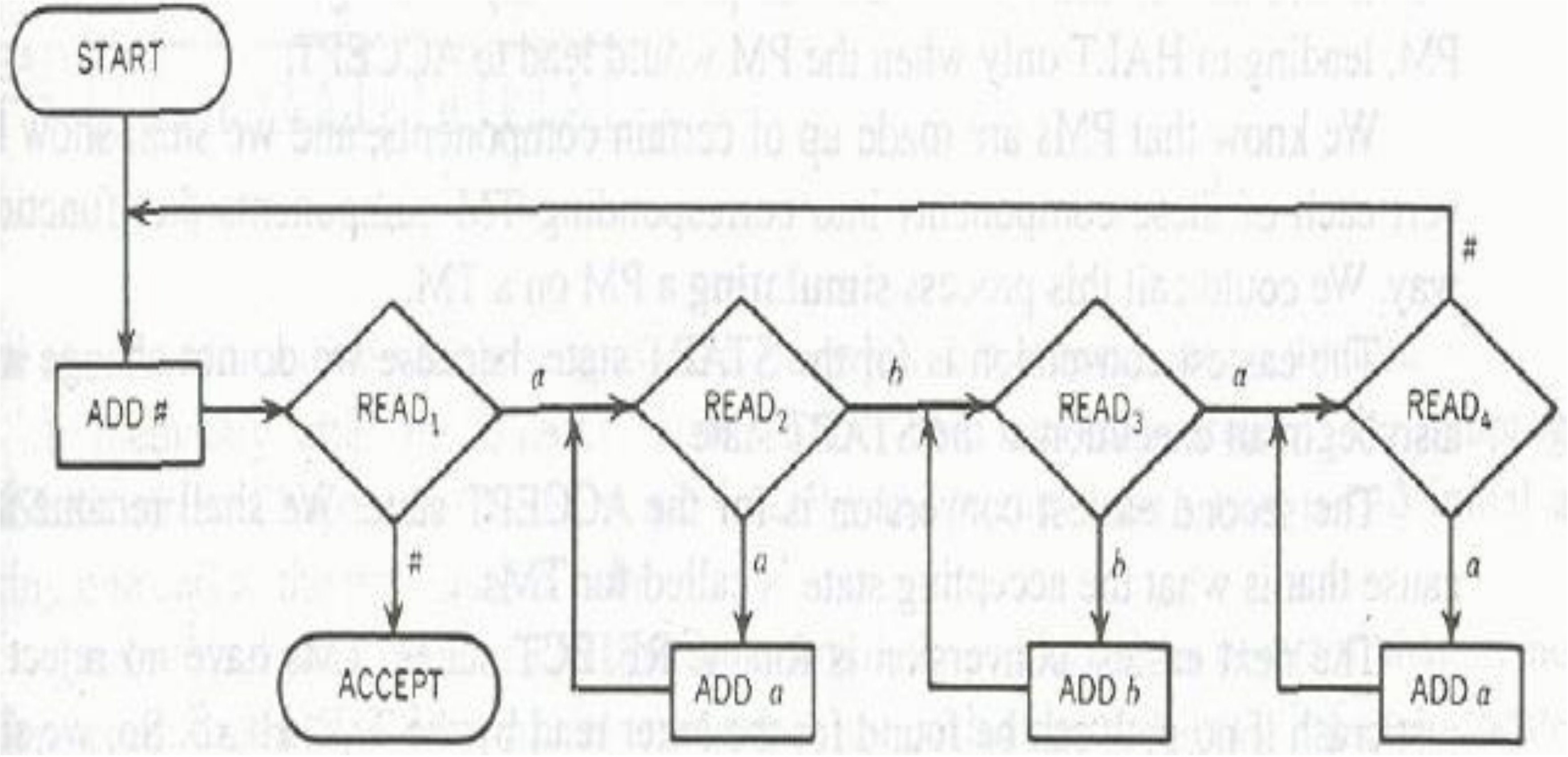
5. **A START** state (un-enterable) and some halt states called **ACCEPT and REJECT**

Example

- Consider the following PM and guess the language?



Another POST Machine



Simulating a PM on TM

- Theorem:

  A language accepted by **PM** can also be accepted by **TM**.

Proof:

We can simulate a PM over TM. PM consists of many components.

By simulating we means, we will show how to convert each of these components into corresponding TM components that function the same way.

Simulating PM on TM

- Start state of PM is same as TM’s start state. So no conversion needed.
- Accept state of PM will be renamed as HALT state of TM.
- REJECT state of PM will be converted by deleting the REJECT states and all the edges coming into REJECT states.
- Now how to keep track of PM store contents when simulating over TM tape?

PM Store on TM

We now describe how we can use the TM TAPE to keep track of the STORE. Suppose the contents of the STORE look like

$$x_1x_2x_3x_4x_5$$

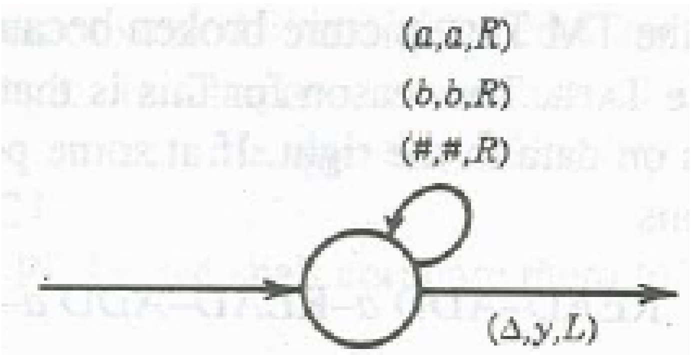
where the x's are from the PM input alphabet Σ or the symbol # and none of them is Δ. We want the corresponding contents of the TM TAPE to be

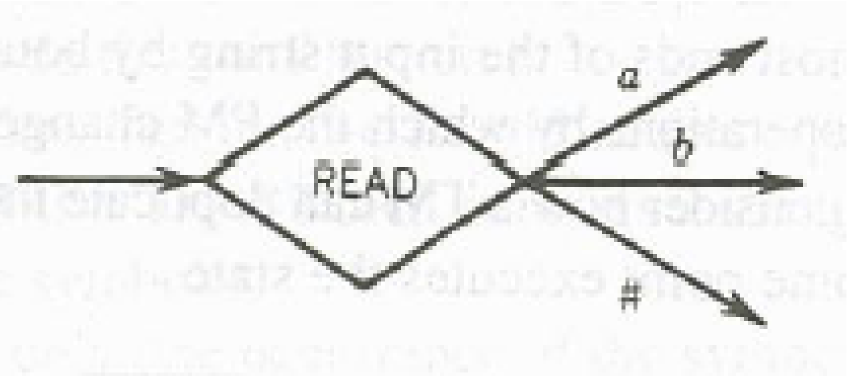
Δ	$\cdots$	Δ	Δ	x_1	x_2	x_3	x_4	x_5	Δ	Δ	$\cdots$

with the TAPE HEAD pointing to one of the x's. Notice that we keep some Δ's on the left of the STORE information, not just on the right, although there will only be finitely many Δ's on the left, since the TAPE ends in that direction

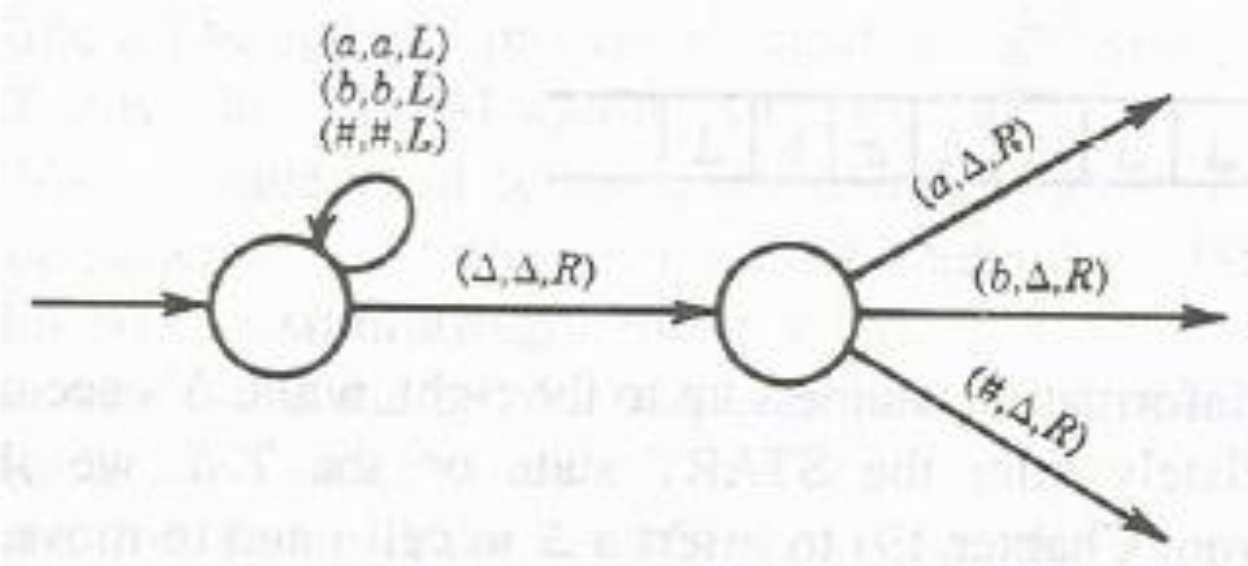
HOW to simulate STORE over TAPE?

- Can be simulated b

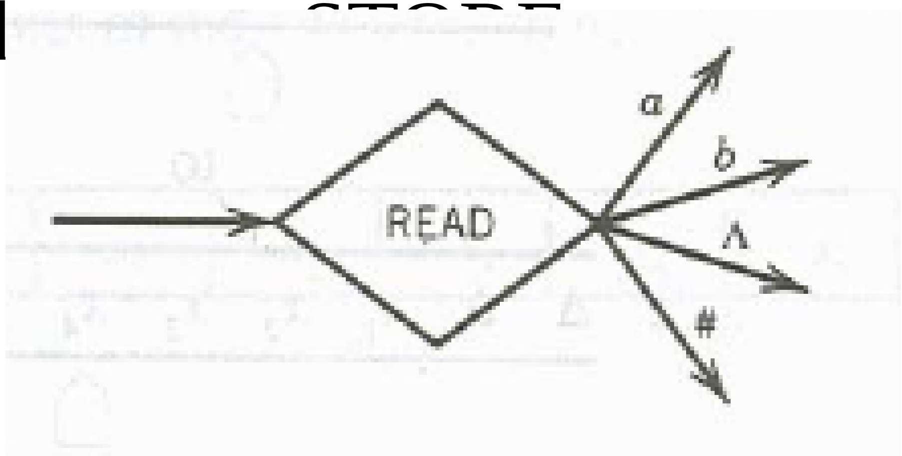




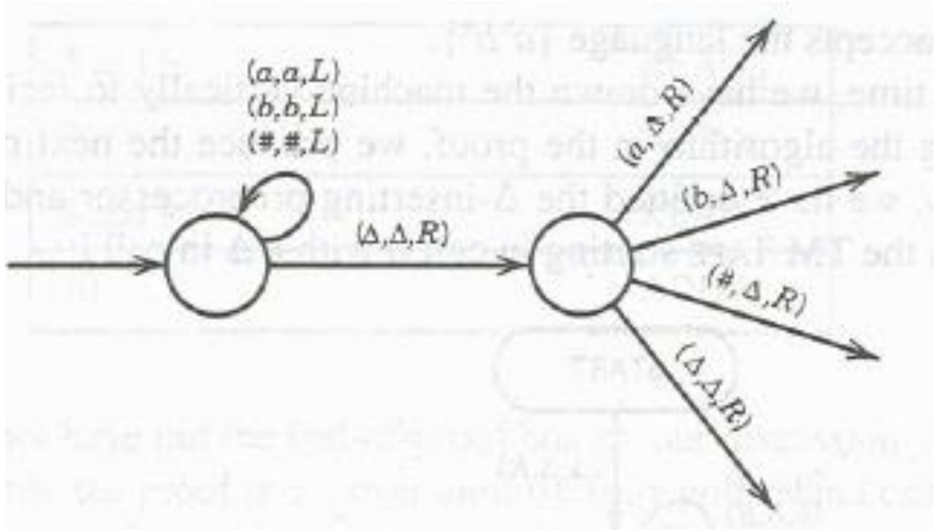
Can be simulated

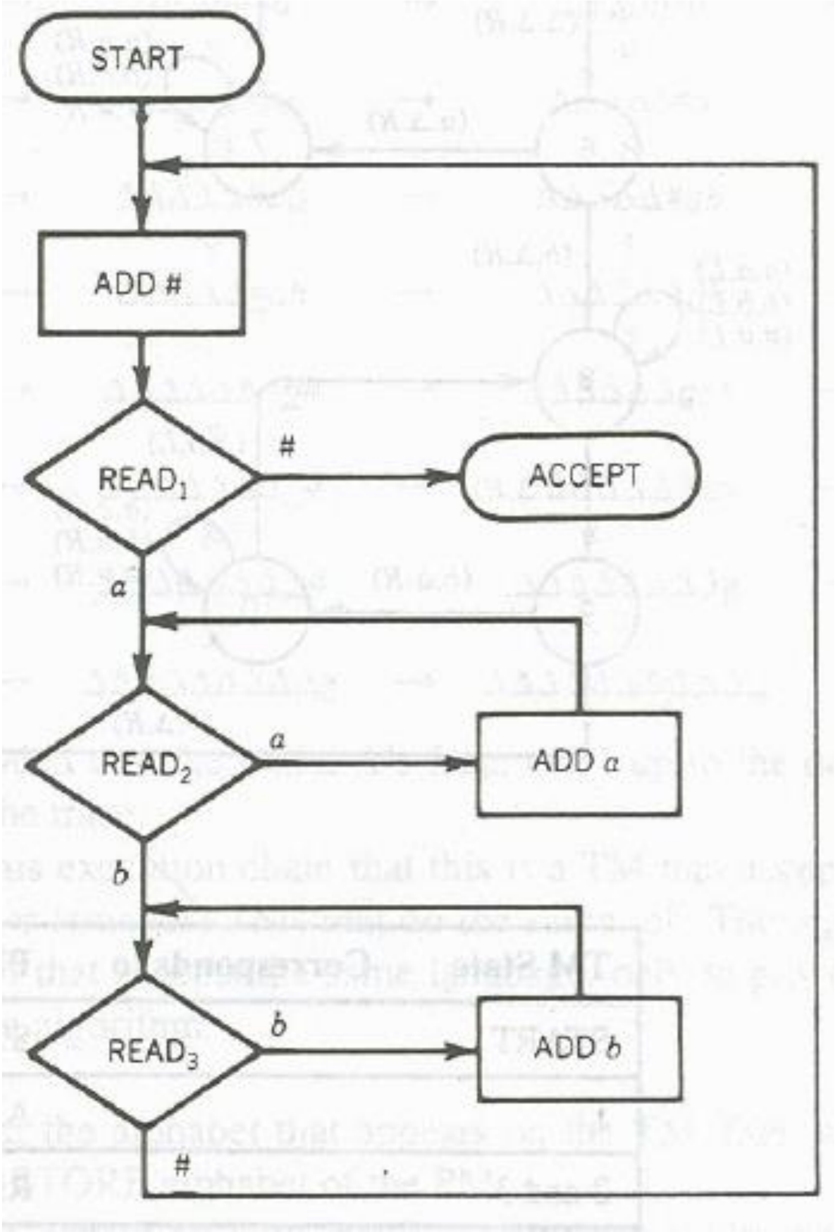


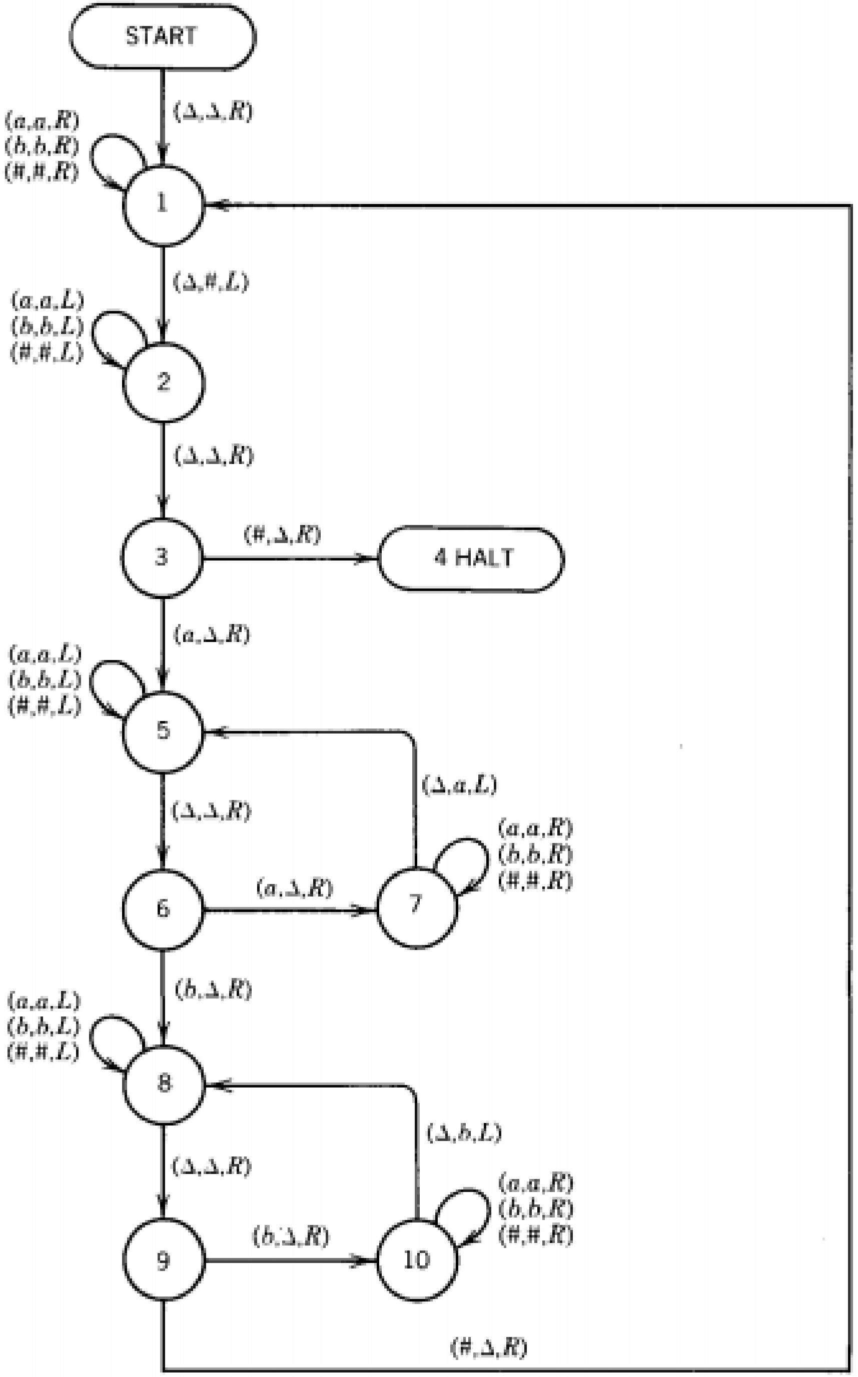
If PM read



Then it becomes




START
ADD #
READ$_1$
#
ACCEPT
a
READ$_2$
a
ADD a
b
READ$_3$
b
ADD b
#


START
(Δ,Δ,R)
(a,a,R)
(b,b,R)
(#,#,R)
1
(Δ,#,L)
(a,a,L)
(b,b,L)
(#,#,L)
2
(Δ,Δ,R)
3
(#,Δ,R)
4 HALT
(a,Δ,R)
(a,a,L)
(b,b,L)
(#,#,L)
5
(Δ,a,L)
(Δ,Δ,R)
(a,a,R)
(b,b,R)
(#,#,R)
6
(a,Δ,R)
7
(b,Δ,R)
(a,a,L)
(b,b,L)
(#,#,L)
8
(Δ,b,L)
(Δ,Δ,R)
(a,a,R)
(b,b,R)
(#,#,R)
9
(b,Δ,R)
10
(#,Δ,R)